Exact reciprocal-simplex product densities below a prescribed spatial scale

Abstract

Spatial earthquake statistics motivate point-process models whose second, third and fourth product densities are proportional to the reciprocals of pair distance, triangle area and tetrahedron volume. Exact identities of this kind cannot hold at every scale for a stationary process of positive intensity. We show that they can hold simultaneously on every nondegenerate configuration below any prescribed diameter, at any independently prescribed positive intensity. The construction chooses between two whole-space laws: a process on separated random spheres and a process of separated clusters containing at most three points. Sphere geometry supplies the exact fourth product density, while the small clusters correct the two lower densities. All factorial expectation measures through order four are absolutely continuous on the whole configuration space, with densities finite almost everywhere. The process is locally finite and has finite third count moments, although its fourth count moment is infinite in every nonempty open ball. The construction is nonergodic.

1 Introduction

Earthquake hypocentres form spatial patterns concentrated around faults and fault systems. A point-process description asks how these patterns appear in the frequencies of pairs, triples and larger collections of events. The work of Kagan and Knopoff on the two-point correlation function [3], followed by Kagan's studies of the three-point and four-point moment functions [4, 5], connects these frequencies with the geometry of the simplex determined by the selected events. Their empirical comparisons suggest approximate reciprocal distance, area and volume relations. In the three-point study, the comparison is with a Poisson catalogue having matched boundaries and depth distribution. The four-point study explicitly discusses a divergent height integral and a lower-height regularisation near the base plane [5, p. 730]. These observations motivate a compatibility question; they do not assert the exact identities that we study below.

Daley and Vere-Jones [2, Exercise 12.6.11] discuss these forms in a question about their compatibility with a stationary point process. Their formulation uses proportional or asymptotic notation; it does not supply the global equality assumptions considered here. We make the domain and the meaning of the densities explicit. We ask for exact identities for the *full* ordered product densities, and we distinguish identities at all distances from identities below a fixed spatial scale. This distinction is necessary: a reciprocal-distance second product density at every distance already contradicts the growth of the mean count in large balls.

Fix a positive intensity λ and a positive diameter M. Our main result constructs a

stationary simple point process on $\mathbb{R}^3$ for which

$$\rho^{(2)}(x_1, x_2) = \frac{a_2}{D(x_1, x_2)},$$
$$\rho^{(3)}(x_1, x_2, x_3) = \frac{a_3}{A(x_1, x_2, x_3)},$$
$$\rho^{(4)}(x_1, x_2, x_3, x_4) = \frac{a_4}{V(x_1, x_2, x_3, x_4)}$$

whenever the corresponding configuration is nondegenerate and has diameter less than M, apart from Lebesgue-null sets. Here D, A and V are ordinary distance, area and volume. The positive constants a_2, a_3, a_4 are supplied by the construction. They are not independently prescribed. There is no lower distance, area, volume or altitude bound, so very thin triangles and almost coplanar tetrahedra remain in the required domain.

The fourth formula has a consequence that is easy to miss. If a fourth vertex approaches the plane of a fixed noncollinear triangle, its reciprocal tetrahedron volume grows as the reciprocal of the normal distance to that plane. Integration in that normal direction diverges. Thus any process satisfying the local fourth identity must have infinite fourth count moments on open balls. This does not prevent a count from being finite almost surely. We use factorial expectation measures as extended nonnegative measures, and prove separately that their densities are finite almost everywhere.

The construction begins with a surprisingly simple description. Toss one fair coin and use its outcome to choose the entire point-process law. The first law places Poisson points on a stationary family of spheres whose surfaces are separated by more than M. Four nearby points must then lie on one sphere. Four affinely independent points determine a unique circumsphere, and the Jacobian of that change of coordinates is proportional to their tetrahedron volume. A suitable distribution of sphere radii cancels the remaining factor, giving the exact fourth identity.

The same calculation gives the first three product densities of the sphere process. The second and third have reciprocal-simplex factors multiplied by Gaussian attenuation terms. An internal length parameter makes their deficits small in the integrals needed for a correction, without changing the intensity or the leading coefficients. The second law uses clusters of at most three points to supply those deficits. Its fourth product density vanishes on configurations of diameter below M. A choice between whole laws then adds the desired product densities with no mixed tuples between the two alternatives.

The spherical change of variables is an instance of the Euclidean Blaschke–Petkantschin formula stated in Chapron [1, equation (1)]. We give the required specialisation and the lower-order calculations. That geometric identity by itself does not establish the existence of the process: the probability budget, the correction kernels and the global regularity arguments are separate parts of the proof. We make no novelty or priority claim for the geometric identity or for the resulting construction.

The example is a realisability result for these exact local product densities. It is nonergodic, and we do not claim that it gives a statistical model fitted to earthquake catalogues. Its purpose is to separate the compatibility question from additional requirements such as ergodicity, finite fourth moments or an empirical fit.

2 Factorial expectation measures and product densities

Let X be a point process on $\mathbb{R}^3$, regarded as a random counting measure. We write $X(B)$ for the number of points in a Borel set B. The process is *locally finite* if $X(B) < \infty$ almost surely for every bounded B, and *simple* if its points have multiplicity one. It is *stationary* if

translating every point by any fixed vector leaves its law unchanged. A stationary process has intensity λ when

$$\mathbb{E}X(B) = \lambda|B| \tag{2.1}$$

for every bounded Borel set B, where $|B|$ is three-dimensional Lebesgue measure. In this paper $0 < \lambda < \infty$.

2.1 Counting ordered tuples

For $n \geq 1$, the nth factorial expectation measure $\alpha_X^{(n)}$ is defined by

$$\int h \, \mathrm{d}\alpha_X^{(n)} = \mathbb{E} \sum_{x_1,\dots,x_n \in X}^{\neq} h(x_1, \dots, x_n), \qquad h \geq 0 \text{ measurable.} \tag{2.2}$$

The superscript $\neq$ means that the selected points are pairwise distinct, and the tuples are ordered. For disjoint sets $B_1, \dots, B_n$, this measure gives $\mathbb{E}\prod_i X(B_i)$ on $B_1 \times \cdots \times B_n$. For a single set B, it gives

$$\alpha_X^{(n)}(B^n) = \mathbb{E}\big(X(B)\big)_n, \qquad (m)_n = m(m-1)\cdots(m-n+1). \tag{2.3}$$

Both sides may be infinite. These are the factorial moment measures in the terminology of [2, Chapter 9]; "expectation measure" here emphasises that local finiteness of the measure is not assumed.

If there is a measurable $\rho_X^{(n)} \geq 0$ such that

$$\alpha_X^{(n)}(\,\mathrm{d}x_1 \cdots \mathrm{d}x_n) = \rho_X^{(n)}(x_1, \dots, x_n)\,\mathrm{d}x_1 \cdots \mathrm{d}x_n, \tag{2.4}$$

we call $\rho_X^{(n)}$ a *product density* or *factorial density*. The equality is an identity of full extended measures. It includes all ordered tuples and leaves no singular measure on collinear, coplanar or other exceptional configurations. We require the density to be finite Lebesgue-almost everywhere, but its integral over a bounded set may be infinite. A finite-almost-everywhere function need not be locally integrable.

At intensity λ, the normalised functions are

$$g_n(x_1, \dots, x_n) = \frac{\rho_X^{(n)}(x_1, \dots, x_n)}{\lambda^n}. \tag{2.5}$$

A homogeneous Poisson process has $\rho^{(n)} = \lambda^n$ and $g_n = 1$. Normalisation changes only the coefficients in the reciprocal-simplex identities. It does not turn a full product density into a cumulant or a connected correlation function. We use full product densities throughout.

2.2 Two elementary constructions

We will use a Poisson process with a random intensity measure. If Λ is a random locally finite nonatomic measure and, conditionally on Λ, Y is a Poisson process with intensity measure Λ, then

$$\alpha_Y^{(n)} = \mathbb{E}\Lambda^{\otimes n}. \tag{2.6}$$

To see this, first use the Poisson factorial identity for products of pairwise disjoint sets, condition on Λ, and then extend to nonnegative measurable functions by simple functions and monotone convergence. The nonatomic assumption makes all product diagonals null. Such a process is called a Cox process. Its directing measure may be singular with respect to volume; in our example it is supported on sphere surfaces. Identity (2.6) is still valid.

A different operation is a mixture of laws. If a coin chooses Y with probability θ and C with probability $1-\theta$, defining the whole process X, then conditioning on that coin gives

$$\alpha_X^{(n)} = \theta\alpha_Y^{(n)} + (1-\theta)\alpha_C^{(n)}. \tag{2.7}$$

This follows directly from (2.2), also when expectations are infinite. A realisation contains points from the selected law only. This differs from superposing two independent processes, whose full product densities also contain tuples using points from both processes.

3 The diameter-capped problem

For points in $\mathbb{R}^3$, put

$$\begin{aligned} D(x_1,x_2) &= |x_2-x_1|, \\ A(x_1,x_2,x_3) &= \frac{1}{2}|(x_2-x_1)\times(x_3-x_1)|, \\ V(x_1,x_2,x_3,x_4) &= \frac{1}{6}\left|\det[x_2-x_1, x_3-x_1, x_4-x_1]\right|. \end{aligned} \tag{3.1}$$

Write $(\Delta_2,\Delta_3,\Delta_4) = (D,A,V)$ and define

$$E_n(M) = \{(x_1,\dots,x_n) : \operatorname{diam}\{x_1,\dots,x_n\} < M,\ \Delta_n(x_1,\dots,x_n) > 0\}. \tag{3.2}$$

The strict diameter bound does not bound the simplex away from degeneracy. For example, triangles in $E_3(M)$ can have arbitrarily large circumradius.

Theorem 3.1. *For every independently prescribed pair $\lambda > 0$ and $M > 0$, there is a stationary, simple, locally finite point process X on $\mathbb{R}^3$ with intensity λ, and finite positive constants a_2, a_3, a_4, such that the full factorial expectation measures of orders $2,3,4$ have globally defined Lebesgue densities finite almost everywhere and*

$$\rho_X^{(n)}(\mathbf{x}) = \frac{a_n}{\Delta_n(\mathbf{x})} \quad \text{for almost every } \mathbf{x}\in E_n(M), \qquad n=2,3,4. \tag{3.3}$$

The constants and the process may depend on (λ, M). The process has finite third count moments on every bounded set and infinite fourth count moments in every nonempty open ball. It is nonergodic under translations.

There is no claim that a prescribed triple (a_2,a_3,a_4) can be realised. Outside the sets $E_n(M)$ the densities are unconstrained, subject to the global measure and finiteness requirements of the theorem. Values of a density on a Lebesgue-null set may be chosen arbitrarily; this permission does not permit a singular component of its measure.

For clarity, let us distinguish this problem, denoted S1, from the unrestricted version S0, in which the three equalities are required on all nondegenerate configurations without a diameter cap. S0 is false.

Proposition 3.2. *No stationary point process of finite positive intensity has full second factorial density $a_2/|x_2-x_1|$ almost everywhere on distinct pairs in $\mathbb{R}^3$, with $0 < a_2 < \infty$ and no other part of the second factorial measure.*

Proof. Let B_R be a ball of radius R and set $N_R = X(B_R)$. The assumed density is integrable on B_R^2, since

$$\int_{B_R}\int_{B_R} \frac{\mathrm{d}y\,\mathrm{d}x}{|x-y|} \le |B_R| \int_{B(0,2R)} \frac{\mathrm{d}u}{|u|} = 8\pi R^2 |B_R|.$$

Thus N_R has finite second moment, and nonnegativity of its variance gives

$$\lambda^2|B_R|^2 \le \mathbb{E}N_R^2 = \lambda|B_R| + \mathbb{E}(N_R)_2 \le \lambda|B_R| + 8\pi a_2 R^2|B_R|.$$

After dividing by $|B_R|^2 = \left(4\pi R^3/3\right)^2$, we obtain $\lambda^2 \le 3\lambda/(4\pi R^3) + 6a_2/R$, which is impossible as $R \to \infty$. □

This obstruction uses no fourth-moment assumption. It requires a change to the unrestricted tail of the second density, but it does not uniquely determine a replacement problem. The common diameter cap in S1 is the particular domain considered in Theorem 3.1.

4 Existence by a mixture of two processes

We begin with the final random choice, since it explains the job of each component. Let Y be a process on separated spheres, and let C be a process of separated clusters containing at most three points. The process we seek has law

$$\mathcal{L}(X) = \frac{1}{2}\mathcal{L}(Y) + \frac{1}{2}\mathcal{L}(C). \tag{4.1}$$

The coin is tossed once for the whole of $\mathbb{R}^3$. Our task is therefore to construct two actual laws whose intensities average to λ and whose full product densities average to (3.3).

The correction process has a bounded intensity capacity because its clusters are separated. We allow the sphere process to supply the rest. Set

$$\kappa = \frac{1}{36e\pi M^3}, \qquad \lambda_C = \min\{\lambda, \kappa/2\}, \qquad \lambda_Y = 2\lambda - \lambda_C. \tag{4.2}$$

Both intensities are strictly positive and their average is λ. The construction below works for arbitrarily large λ_Y, while the correction uses $\lambda_C \le \kappa/2$.

4.1 Process 1: Poisson points on separated spheres

Let

$$C_0 = \frac{\pi^{3/2}}{3\sqrt{3}}. \tag{4.3}$$

Choose a finite $K > 0$ sufficiently large that, with

$$c = \sqrt{\frac{C_0 K^3}{\lambda_Y}}, \qquad F = \frac{K^4}{c^3} = \frac{\lambda_Y^{3/2}}{C_0^{3/2}\sqrt{K}}, \qquad FM \le 10^{-4}. \tag{4.4}$$

For example, $K = 10^8 M^2 \lambda_Y^3 / C_0^3$ satisfies the last inequality with equality. The parameter $L \ge M$ is left free for now. Define

$$f(q) = \frac{K}{L}e^{-q/L^2}, \qquad a(R) = \frac{c}{f(R^2)}, \qquad \gamma(R) = \frac{2Rf(R^2)^4}{c^3} = \frac{2FR}{L^4}e^{-4R^2/L^2}. \tag{4.5}$$

We will construct a stationary random family of spheres whose first intensity on centre–radius space is $\gamma(R)\,\mathrm{d}z\,\mathrm{d}R$. On a sphere with centre z and radius R, the Poisson surface intensity will be $a(R)/(2R)$. These choices make the fourth-order weight constant after putting $q = R^2$; the calculation appears in Section 4.2.

The required spheres cannot be chosen independently, because nearby points would then also come from different spheres. Instead, choose a single radius band and arrange its surfaces far enough apart. Put

$$H = 2M, \qquad b_k = 2^k M, \qquad P_k = 4b_k + 2M, \qquad k = 0, 1, \dots, \tag{4.6}$$

and partition the positive radii into

$$I_0 = (0, M], \qquad I_k = (b_k/2, b_k] \quad (k \geq 1).$$

Define the band probabilities by

$$p_k = HP_k^3 \sup_{R \in I_k} \gamma(R). \tag{4.7}$$

They obey $\sum_k p_k < 0.9$ for every $L \geq M$, as we verify in Lemma 4.1. With probability p_k, select band k; with the remaining probability, select an empty sphere family.

Conditionally on selecting band k, perform the following steps. Choose Z uniformly on $[0, P_k)^3$ and use the centres $Z + P_k m$, $m \in \mathbb{Z}^3$. Independently choose U uniformly on $[0, H)$ and, at each centre, use the candidate radii $R = U + jH \in I_k$, $j \in \mathbb{Z}$. Retain each candidate sphere, independently conditionally on (k, Z, U), with probability

$$\theta_k(R) = \frac{\gamma(R) H P_k^3}{p_k} \in [0, 1]. \tag{4.8}$$

Each realised band contains only finitely many candidate radial indices at each centre. A radius zero, which has phase probability zero, is omitted. Independent marks on the countable set of centre and radius indices specify this random family.

Lemma 4.1 (Probability budget)**.** *The probabilities in* (4.7) *satisfy*

$$\sum_{k \geq 0} p_k < 9000FM \leq 0.9$$

uniformly over $L \geq M$*.*

Proof. Since $P_k \leq 6b_k$, the first band satisfies

$$p_0 \leq 864FM(M/L)^4 \leq 864FM.$$

For $k \geq 1$, use $R \leq b_k$ and $R > b_k/2$ in (4.5). With $t_k = b_k/L$, this gives

$$p_k \leq 864FM\, t_k^4 e^{-t_k^2}.$$

The t_k form a dyadic sequence. For the terms $t_k \leq 1$, the sum of t_k^4 is at most $16/15$. For $t > 1$, the bound $e^{t^2} \geq t^6/6$ gives $t^4 e^{-t^2} \leq 6/t^2$; the sum over the remaining dyadic terms is at most 8. Therefore

$$\sum_k p_k \leq 864FM \left(1 + \frac{16}{15} + 8\right) = \frac{130464}{15} FM < 9000FM.$$

The bound contains no further dependence on L. □

Lemma 4.2 (Sphere family)**.** *The retained sphere family is stationary, its surfaces are pairwise separated by more than* M*, and, for every nonnegative measurable* h*,*

$$\mathbb{E} \sum_{(z,R)\ retained} h(z, R) = \int_{\mathbb{R}^3} \int_0^\infty h(z, R) \gamma(R)\, \mathrm{d}R\, \mathrm{d}z. \tag{4.9}$$

Proof. Uniform translation of the lattice gives, for nonnegative g,

$$\mathbb{E}_Z \sum_{m\in\mathbb{Z}^3} g(Z + P_k m) = P_k^{-3} \int_{\mathbb{R}^3} g(z)\,\mathrm{d}z.$$

Uniform radial phase similarly gives

$$\mathbb{E}_U \sum_{j:\,U+jH\in I_k} g(U + jH) = H^{-1} \int_{I_k} g(R)\,\mathrm{d}R.$$

Multiplying these identities by $p_k\theta_k(R)$, and summing over the bands, proves (4.9). Only the first intensity is asserted; the sphere parameters are dependent.

At a common centre the surfaces of two candidate spheres have distance at least $H = 2M$. At different centres in band k, their distance is at least

$$P_k - 2b_k = 2b_k + 2M > M. \tag{4.10}$$

These bounds survive retention. Finally, translating space changes Z modulo $P_k\mathbb{Z}^3$ and relabels the centre indices. The conditional retention law is unchanged by that relabelling, while the law of Z is uniform. This proves stationarity within each band and for their mixture. □

Write $S(z, R)$ for the sphere and $\sigma_{z,R}$ for its unnormalised surface area measure. Given the retained family, place independent Poisson processes on its spheres, with intensity measures

$$\frac{a(R)}{2R}\sigma_{z,R}. \tag{4.11}$$

Equivalently, sample a Poisson number with mean $2\pi a(R)R$ on each retained sphere and sample its directions independently and uniformly on S^2. Their union is the process Y. Its directing measure is

$$\Lambda = \sum_{(z,R)\text{ retained}} \frac{a(R)}{2R}\sigma_{z,R}. \tag{4.12}$$

The construction already gives an actual stationary, simple and locally finite process. To check local finiteness, condition on the selected band. Only centres within distance b_k of a bounded observation set can affect it, and there are finitely many such centres and candidate radii. Each sphere carries a finite Poisson count. The surface measures have no atoms, and different surfaces are disjoint, so the union is simple. Countable exhaustion by balls makes these statements simultaneous over the whole space. The spatially invariant Poisson draws preserve stationarity.

We will compute the local product densities first. The fact that these are globally well-defined densities requires attention to tuples lying on several spheres, even though such tuples do not occur in $E_n(M)$. The full proof is in Appendix A. It establishes the following facts, which will also be used at the end of the construction.

Proposition 4.3. *For the process Y, the full factorial expectation measures through order four are absolutely continuous on the whole configuration space and have densities finite almost everywhere. Moreover, $\mathbb{E}Y(B)^3 < \infty$ for every bounded Borel set B.*

4.2 Sphere geometry and the first four densities

Four noncoplanar points determine a unique sphere. The corresponding change of coordinates is particularly useful here. Let $\omega_i \in S^2$ and $x_i = z + R\omega_i$, $i = 1, \dots, 4$. The Euclidean spherical formula stated in [1, equation (1)], specialised to dimension three, is

$$\mathrm{d}x_1 \cdots \mathrm{d}x_4 = 6V(\omega_1, \dots, \omega_4) R^8 \, \mathrm{d}z \, \mathrm{d}R \prod_{i=1}^{4} \mathrm{d}\omega_i, \tag{4.13}$$

where $\mathrm{d}\omega$ is surface area measure on the unit sphere and V is the ordinary tetrahedron volume defined in (3.1). Since $V(x_1, \dots, x_4) = R^3 V(\omega_1, \dots, \omega_4)$, the Jacobian is $6VR^5$. Chapron attributes the Euclidean identity to Miles; we use the displayed formula in Chapron, without making a separate attribution claim based on the original Miles paper.

A family with first intensity $\gamma(R) \, \mathrm{d}z \, \mathrm{d}R$, carrying Poisson surface intensity $t(R)\sigma_{z,R}$, therefore has same-sphere fourth density

$$\frac{\gamma(R_*) t(R_*)^4 R_*^3}{6V}, \tag{4.14}$$

evaluated at the unique circumradius R_*. This follows by dividing the parameter weight $\gamma t^4 R^8$ by the Jacobian $6VR^5$. The formula is ordered: no division by 4! is present.

For our process, separation makes this same-sphere contribution the full local fourth density. We also need the lower orders, for which the set of possible sphere centres has positive dimension. The next lemma records all four calculations in a common convention.

Lemma 4.4 (Sphere integrals)**.** *Let $w_n : (0, \infty) \to [0, \infty]$ be measurable. Consider the measure obtained by integrating*

$$w_n(q) \prod_{i=1}^{n} \frac{\sigma_{z,\sqrt{q}}(\, \mathrm{d}x_i)}{2\sqrt{q}}$$

over $z \in \mathbb{R}^3$ and $q > 0$. On nondegenerate configurations its density is

$$r_1 = 2\pi \int_0^\infty \sqrt{q} \, w_1(q) \, \mathrm{d}q, \tag{4.15}$$

$$r_2(x_1, x_2) = \frac{\pi}{2D} \int_{D^2/4}^\infty w_2(q) \, \mathrm{d}q, \tag{4.16}$$

$$r_3(x_1, x_2, x_3) = \frac{1}{8A} \int_{\mathbb{R}} w_3(R_\triangle^2 + h^2) \, \mathrm{d}h, \tag{4.17}$$

$$r_4(x_1, x_2, x_3, x_4) = \frac{w_4(R_*^2)}{48V}. \tag{4.18}$$

Here $R_\triangle$ is the triangle circumradius and R_* is the tetrahedron circumradius. The formulas hold as nonnegative measure identities; local integrability is not assumed.

Proof. The one-dimensional radial coarea formula identifies

$$\frac{\sigma_{z,\sqrt{q}}(\, \mathrm{d}x)}{2\sqrt{q}} = \delta(|x - z|^2 - q) \, \mathrm{d}x.$$

We use the right-hand side only as notation for the measure on the left. Thus the density is obtained by integrating

$$w_n(q) \prod_{i=1}^{n} \delta(|x_i - z|^2 - q) \tag{4.19}$$

over (z, q). We justify this representation after computing the constraint Jacobians.

For one point, translating x_1 to zero and using polar coordinates in z gives $\int_{\mathbb{R}^3} \delta(|z|^2 - q)\,\mathrm{d}z = 2\pi\sqrt{q}$. For two points, write $x_1 = 0$ and $x_2 = de$, with $|e| = 1$. Subtracting the first constraint from the second gives $d^2 - 2d(z \cdot e) = 0$. It fixes $z \cdot e = d/2$ with Jacobian $2d$. The first constraint fixes $q = d^2/4 + |z_\perp|^2$ with Jacobian one. Integrating the remaining two centre coordinates in polar form gives

$$\frac{1}{2d} \int_{\mathbb{R}^2} w_2(d^2/4 + |v|^2)\,\mathrm{d}v = \frac{\pi}{2d} \int_{d^2/4}^{\infty} w_2(q)\,\mathrm{d}q.$$

For three points, translate the first to zero and write the others as u, v. The two subtracted constraints fix the projection of the centre onto the triangle plane at its circumcentre. Their Jacobian in orthonormal plane coordinates is $4|u \times v| = 8A$. If h is the remaining normal coordinate, the first constraint fixes $q = R_\triangle^2 + h^2$. This gives (4.17).

For four noncoplanar points, the four constraints determine (z, q) uniquely, with $q = R_*^2 > 0$. Subtracting the first row from the other three in the derivative with respect to (z, q) gives absolute determinant

$$8\,|\det[x_2 - x_1, x_3 - x_1, x_4 - x_1]| = 48V. \tag{4.20}$$

This proves (4.18), consistently with (4.13) after the change $q = R^2$.

For completeness, these calculations can be read as ordinary changes of variables on the incidence set $|x_i - z|^2 = q$. Work first in a compact regular coordinate chart, with q bounded away from zero and with a continuous test function. Replace each displayed δ by a smooth one-dimensional approximate identity. Integrating first in the radial coordinate of each x_i gives the surface measure divided by $2\sqrt{q}$. Alternatively, keep the x_i fixed and use q and $n - 1$ centre coordinates as the constraint coordinates. The nonzero determinants just computed give (4.15)–(4.18) in the limit. The approximate-identity passage is the usual one-dimensional substitution on these regular charts, with compact support and continuous Jacobians. A countable chart decomposition proves equality of the resulting Borel measures, and nonnegative exhaustion extends it to arbitrary measurable weights and tests. All subsequent integrations use Tonelli's theorem. No product of unspecified distributions, or finite fourth moment, is needed. □

For Y, every tuple of diameter below M lies on a single retained sphere. By (2.6) and (4.9), its full local factorial measure is therefore

$$\int_{\mathbb{R}^3} \int_0^{\infty} \gamma(R) \left(\frac{a(R)}{2R}\right)^n \sigma_{z,R}^{\otimes n}(\,\mathrm{d}\mathbf{x})\,\mathrm{d}R\,\mathrm{d}z. \tag{4.21}$$

For $n = 1$ the same expression gives the full first measure everywhere. Putting $q = R^2$, the weight in Lemma 4.4 becomes

$$w_n(q) = \frac{\gamma(\sqrt{q})a(\sqrt{q})^n}{2\sqrt{q}}, \qquad (w_1, w_2, w_3, w_4) = \left(\frac{f^3}{c^2}, \frac{f^2}{c}, f, c\right). \tag{4.22}$$

Define

$$A_2 = \frac{\pi K^2}{4c}, \qquad A_3 = \frac{K\sqrt{\pi}}{8}. \tag{4.23}$$

The four densities are

$$\rho_Y^{(1)} = \frac{2\pi}{c^2}\int_0^\infty \sqrt{q}\, f(q)^3\,\mathrm{d}q = \frac{C_0 K^3}{c^2} = \lambda_Y, \tag{4.24}$$

$$\rho_Y^{(2)} = \frac{A_2}{D} e^{-D^2/(2L^2)} \qquad \text{on } E_2(M), \tag{4.25}$$

$$\rho_Y^{(3)} = \frac{A_3}{A} e^{-R_\triangle^2/L^2} \qquad \text{on } E_3(M), \tag{4.26}$$

$$\rho_Y^{(4)} = \frac{c}{48V} \qquad \text{on } E_4(M). \tag{4.27}$$

For the first line,

$$\int_0^\infty q^{1/2} e^{-3q/L^2}\,\mathrm{d}q = L^3\sqrt{\pi}/(6\sqrt{3}).$$

The pair integral uses

$$\int_{D^2/4}^\infty e^{-2q/L^2}\,\mathrm{d}q = (L^2/2)e^{-D^2/(2L^2)},$$

and the triangle integral uses $\int_{\mathbb{R}} e^{-h^2/L^2}\,\mathrm{d}h = L\sqrt{\pi}$. Thus λ_Y, A_2, A_3, c are independent of the still-free scale L.

The fourth density is already exactly of the desired form. After the fair coin it will have coefficient $c/96$. The lower two densities fall short of the reciprocal forms because of their exponential factors. Their pointwise deficits are nonnegative. We need more than pointwise smallness: the triangle deficit contributes pairs when realised by three-point clusters, so its pair marginal must be controlled uniformly, including for thin triangles. That is the next step.

4.3 The two deficits and their marginal bound

For a translation-invariant kernel h of j points, write

$$(\mathcal{M}h)(x_1,\dots,x_{j-1}) = \int_{\mathbb{R}^3} h(x_1,\dots,x_j)\,\mathrm{d}x_j.$$

Its *anchored integral* is the integral with the first point fixed at zero and the other points integrated over $\mathbb{R}^3$. Fixing one point removes the infinite factor arising from translation invariance. Set $K_2 = \mathbf{1}_{E_2(M)}/D$, with value zero on degenerate pairs, and put

$$B_L = \frac{8\pi A_3 M^3}{L}, \qquad \delta = 2B_L. \tag{4.28}$$

Define the desired correction densities

$$q_3 = \frac{A_3 \mathbf{1}_{E_3(M)}}{A}\left(1 - e^{-R_\triangle^2/L^2}\right), \tag{4.29}$$

$$q_2 = \frac{\mathbf{1}_{E_2(M)}}{D}\left[\delta + A_2\left(1 - e^{-D^2/(2L^2)}\right)\right]. \tag{4.30}$$

Both kernels are symmetric, translation invariant and nonnegative, and we set them to zero on degenerate configurations. The extra term δ/D in the pair correction allows for pairs that must accompany a three-point cluster. It changes the eventual coefficient a_2, which the problem leaves free.

Lemma 4.5 (Uniform marginal bound)**.** *For every $L \geq M$,*

$$\mathcal{M}q_3 \leq B_L K_2. \tag{4.31}$$

Consequently, the kernels

$$F_3 = q_3, \qquad F_2 = q_2 - \mathcal{M}q_3 \tag{4.32}$$

are nonnegative, with $F_2 \geq B_L K_2$. Their anchored integrals are finite and strictly positive.

Proof. Fix a pair $0, de$, with $0 < d < M$ and $|e| = 1$. For the third vertex z, let r be its distance from the line through the pair. The triangle area is $dr/2$, and its circumradius is

$$R_\triangle = \frac{|z|\,|z - de|}{2r} \leq \frac{M^2}{2r} \quad \text{on } E_3(M). \tag{4.33}$$

Let $\varepsilon = M^2/(2L) \leq M/2$. Since $1 - e^{-t} \leq \min\{1, t\}$ for $t \geq 0$, we have

$$\begin{aligned} d \int_{\mathbb{R}^3} q_3(0, de, z)\,\mathrm{d}z &\leq 2A_3 \int_{B(0,M)} \frac{1}{r} \min\{1, \varepsilon^2/r^2\}\,\mathrm{d}z \\ &\leq 8\pi A_3 M \int_0^M \min\{1, \varepsilon^2/r^2\}\,\mathrm{d}r \\ &\leq 16\pi A_3 M \varepsilon = B_L. \end{aligned}$$

In the second line, cylindrical coordinates cancel the factor $1/r$, and the axial length is at most $2M$. For the last line, split the integral at ε; it is at most 2ε. Collinear third vertices form a null set. If the pair lies outside $E_2(M)$, no triangle in the support of q_3 contains it, so the bound also holds there.

Let Q_j be the anchored integral of q_j and Z_j that of F_j. The pair integral $\int K_2(0, u)\,\mathrm{d}u = 2\pi M^2$, together with $1 - e^{-d^2/(2L^2)} \leq d^2/(2L^2)$, gives

$$Q_2 \leq \frac{32\pi^2 A_3 M^5}{L} + \frac{\pi A_2 M^4}{2L^2}. \tag{4.34}$$

Indeed, the extra term integrates to $2\pi M^2 \delta$, and the other term is bounded by

$$4\pi A_2 \int_0^M d^3/(2L^2)\,\mathrm{d}d.$$

Integrating (4.31) proves $Q_3 < \infty$. No infinite quantities are subtracted in (4.32). The lower bound $q_2 \geq 2B_L K_2$ yields $F_2 \geq B_L K_2$, and Tonelli gives

$$Z_3 = Q_3 > 0, \qquad Z_2 = Q_2 - Q_3 > 0. \tag{4.35}$$

The strict positivity of Q_3 follows from the strictly positive circumradius on every nondegenerate triangle; that of Z_2 follows from its displayed lower bound. □

We can now choose the sphere scale. Take

$$L = \max\left\{M,\ \frac{128\pi^2 A_3 M^5}{\lambda_C},\ M^2\sqrt{\frac{2\pi A_2}{\lambda_C}}\right\}. \tag{4.36}$$

Each of the two summands in (4.34) is then at most $\lambda_C/4$, so

$$Q_2 \leq \lambda_C/2. \tag{4.37}$$

All quantities on the right of (4.36) were fixed before L. Lemma 4.1 remains valid at this value because its probability bound was uniform in $L \geq M$.

The triangle deficit need not be uniformly small over shapes: $R_\triangle$ is unbounded on $E_3(M)$. The estimate we use is instead the uniform pair-marginal bound (4.31). This is why no lower altitude cutoff enters the argument.

4.4 Process 2: separated finite correction clusters

We next realise q_2 and q_3 as the full local product densities of one process with no eligible four-point tuples. Let Π be a homogeneous Poisson process of centres with intensity

$$\eta = \frac{1}{36\pi M^3}. \tag{4.38}$$

Retain a centre if there is no other centre within distance $3M$. The retained centres are dependent, but have separation greater than $3M$ almost surely and intensity

$$\eta \exp\{-\eta |B(0,3M)|\} = \eta e^{-1} = \kappa. \tag{4.39}$$

Here $|B(0,3M)| = 36\pi M^3$.

The first-intensity calculation can be obtained directly by Poisson insertion. For a nonnegative measurable test h,

$$\mathbb{E} \sum_{z \in \Pi} h(z, \Pi \setminus \{z\}) = \eta \int_{\mathbb{R}^3} \mathbb{E} h(z, \Pi)\, \mathrm{d}z. \tag{4.40}$$

On a bounded region, condition on the Poisson count, select one point, and use $n/n! = 1/(n-1)!$ in the resulting nonnegative series. Exhaustion proves (4.40) in the whole space. Applying it to the empty-ball condition gives (4.39) by the Poisson void probability. Pairs at exactly distance $3M$ have zero expected count in bounded windows, so strict separation holds almost surely.

Set

$$\beta = \lambda_C - Q_2 + \frac{Q_3}{2} = \lambda_C - Z_2 - \frac{Z_3}{2}. \tag{4.41}$$

By (4.37), $\beta \geq \lambda_C/2 > 0$. Independently at each retained centre, choose a cluster size in $\{0,1,2,3\}$ with probabilities

$$\pi_1 = \frac{\beta}{\kappa}, \qquad \pi_2 = \frac{Z_2}{2\kappa}, \qquad \pi_3 = \frac{Z_3}{6\kappa}, \qquad \pi_0 = 1 - \pi_1 - \pi_2 - \pi_3. \tag{4.42}$$

These are valid probabilities because

$$\kappa(\pi_1 + \pi_2 + \pi_3) = \lambda_C - \frac{Q_2}{2} + \frac{Q_3}{6} \leq \lambda_C \leq \kappa. \tag{4.43}$$

We used $0 < Q_3 < Q_2$ in the inequality.

A size-zero cluster emits nothing and a size-one cluster emits its centre. For size $j = 2, 3$, emit the centre z and $j-1$ offsets $z + y_2, \ldots, z + y_j$, with joint density

$$\frac{F_j(0, y_2, \ldots, y_j)}{Z_j} \quad \text{on } (\mathbb{R}^3)^{j-1}. \tag{4.44}$$

Lemma 4.5 proves that these are probability densities. Every cluster has diameter below M and contains its centre. The union of the emitted points is C.

Points from different clusters are at distance greater than $3M - 2M = M$. Thus every tuple in $E_n(M)$ comes from one cluster. The process is stationary because the thinning rule commutes with translations and the offset laws do not depend on the centre location. It is simple and locally finite: only Poisson centres within distance M of a bounded observation set can contribute, each emits at most three points, and the continuous offset laws give no coincidences.

Proposition 4.6 (The four correction densities)**.** *The process C has intensity λ_C and, on the indicated local domains, full product densities*

$$\rho_C^{(1)} = \lambda_C, \tag{4.45}$$

$$\rho_C^{(2)} = F_2 + \mathcal{M}F_3 = q_2 \qquad \text{on } E_2(M), \tag{4.46}$$

$$\rho_C^{(3)} = F_3 = q_3 \qquad \text{on } E_3(M), \tag{4.47}$$

$$\rho_C^{(4)} = 0 \qquad \text{on } E_4(M). \tag{4.48}$$

Its full factorial expectation measures through order four have globally finite-almost-everywhere densities, and its bounded-set count moments are finite at every order.

Proof. Consider a cluster of size j and an ordered selection of $n \le j$ of its points. After integrating its centre over space, symmetry and translation invariance of F_j make every ordered selection have the same density. More explicitly, permute the selected point labels, choose one selected point as the new anchor, and subtract its offset from the remaining offsets. This linear change of variables has absolute Jacobian one. The kernel is unchanged by the permutation and common translation. Integrating the $j-n$ unused coordinates gives $\mathcal{M}^{j-n}F_j$. There are $j!/(j-n)!$ ordered selections.

The retained-centre intensity κ is sufficient for this calculation. For pairs, size-two clusters contribute $2\kappa\pi_2 F_2/Z_2 = F_2$, and size-three clusters contribute $6\kappa\pi_3\mathcal{M}F_3/Z_3 = \mathcal{M}F_3$. For triples, the contribution is $6\kappa\pi_3 F_3/Z_3 = F_3$. There is no ordered selection of four points from one cluster. Since separation excludes other local allocations, these are the full local densities in (4.46)–(4.48). The first density is

$$\kappa(\pi_1 + 2\pi_2 + 3\pi_3) = \beta + Z_2 + Z_3/2 = \lambda_C.$$

This calculation does not treat the retained centres as independent.

We still need the full measures outside the local domains. Give the same independent marks to every original Poisson centre before retention, and let $C^\sharp$ be the unthinned emitted process. Then $C \subseteq C^\sharp$ and, for bounded B,

$$C^\sharp(B) \le 3\Pi(B^{+M}), \qquad B^{+M} = \{x : \operatorname{dist}(x, B) < M\}. \tag{4.49}$$

The right-hand side has finite moments of all orders. Absolute continuity can be checked by retaining every parent allocation in the unthinned process. Put $F_1 = Z_1 = 1$, and for $1 \le n \le 3$ define the same-parent kernel

$$G_n = \eta \sum_{j=n}^{3} \frac{\pi_j j!}{(j-n)! Z_j} \mathcal{M}^{j-n} F_j, \qquad G_n = 0 \quad (n > 3). \tag{4.50}$$

In particular $G_1 = \eta \sum_{j=1}^{3} j\pi_j$ is a constant. Label a tuple by the partition $\mathcal{P}$ of its indices according to its distinct Poisson parents. The Poisson factorial identity for those parents, and the ordered-selection calculation above, give

$$\rho_{C^\sharp}^{(n)}(\mathbf{x}) = \sum_{\mathcal{P}} \prod_{I \in \mathcal{P}} G_{|I|}(x_I). \tag{4.51}$$

The sum is over every set partition of $\{1, \dots, n\}$. The identity follows from iterated insertion (4.40); all summands are nonnegative, so Tonelli applies. This proves global absolute continuity, including all different-parent tuples. The moment bound (4.49) makes the full measure locally finite, so its density is finite almost everywhere.

For nonnegative tuple tests, the coupling gives $\alpha_C^{(n)} \le \alpha_{C^\sharp}^{(n)}$. Hence C inherits global absolute continuity and finite-almost-everywhere densities. It also inherits the moment bounds from (4.49). □

4.5 The final mixture

All the ingredients now exist at finite parameter values. The final step is short because the two components were designed for the same coin.

Corollary 4.7. *The law* (4.1) *satisfies Theorem 3.1, with*

$$a_2 = \frac{A_2 + \delta}{2}, \qquad a_3 = \frac{A_3}{2}, \qquad a_4 = \frac{c}{96}. \tag{4.52}$$

Its normalised functions on $E_n(M)$ *are* $g_n = a_n/(\lambda^n \Delta_n)$.

Proof. Both alternatives are stationary, simple and locally finite, and their mean intensities average to λ by (4.2). Proposition 4.3 and Proposition 4.6 give globally finite-almost-everywhere full densities. A finite mixture preserves these properties, and (2.7), (4.25)–(4.27) and (4.46)–(4.48) give exactly (4.52).

The parameter choices have no circular dependence. First (λ, M) determines $(\kappa, \lambda_C, \lambda_Y)$. Next choose K, c by (4.4), then obtain A_2, A_3 from (4.23). Choose L from (4.36), and finally define B_L, δ, the sphere band law and the cluster kernels. The uniform sphere budget and (4.37) verify both probability laws at that L. This works for every prescribed pair, without a restriction on λM^3.

Finite third count moments follow from the corresponding component bounds. The fourth-moment assertion is proved just below. Finally, the sphere law is empty with probability at least $1 - \sum_k p_k > 0.1$, so $\mathbb{P}(X = \varnothing) > 0$. Positive intensity implies $\mathbb{P}(X \neq \varnothing) > 0$. The event that the whole configuration is empty is measurable and invariant under every translation, with probability strictly between zero and one. Thus X is nonergodic. □

Proposition 4.8. *Any process with the full local fourth density* a_4/V *on* $E_4(M)$*, where* $a_4 > 0$*, has infinite fourth count moment in every nonempty open ball.*

Proof. Inside an arbitrary nonempty open ball, choose a smaller ball B whose diameter is less than M. Choose three small neighbourhoods of the vertices of a fixed noncollinear triangle in its interior. Shrink them so that every resulting base triangle has area bounded above and away from zero and remains a positive distance from the boundary of B. They form a positive-measure family of base triangles.

For each such triangle, a small planar disc about its barycentre and a uniformly short normal interval remain inside B. Write the fourth vertex in these coordinates as (v, h). The tetrahedron volume is $V = A|h|/3$, while the coordinate Jacobian is one. Apart from $h = 0$, all these configurations are in $E_4(M)$. The integral of a_4/V over the fourth vertex contains a positive multiple of $\int_0^\varepsilon \mathrm{d}h/h = \infty$. Tonelli, followed by integration over the positive-measure family of base triangles, gives $\alpha_X^{(4)}(B^4) = \infty$. Since $(X(B))_4 \leq X(B)^4$, the raw fourth moment is infinite as well. □

The same argument excludes the additional requirement of finite fourth moments even in the capped problem. It does not exclude local finiteness, and it does not impose any further geometric cutoff on the process we have constructed.

5 Simulation of the construction

We now test the local correlation calculation by sampling point coordinates from the construction. We use $M = 1$ and $\lambda = 6 \times 10^{-6}$, with the explicit parameter choices in the proof. These give $\lambda_C = \lambda_Y = \lambda$, $L = 1$, $K \simeq 1.75519 \times 10^{-8}$ and $c \simeq 9.82728 \times 10^{-10}$. The small intensity makes direct sampling instructive, but inefficient. In five million independent

realisations restricted to the ball $B(0, 0.45)$, we observed seven points, each in a different realisation. Four came from the sphere alternative and three from the correction alternative. The expected total point count is 11.4511. Thus no pair, triangle or tetrahedron was observed in these ordinary windows. We keep these data separate from the importance-sampled experiment below.

Exact simulation in a small ball. The separation in the construction means that a ball of diameter less than M meets at most one retained sphere and receives points from at most one correction cluster. Its restricted law is therefore determined by the first measure of the possible intersecting object. For the sphere component, this measure is $\gamma(R)\,dz\,dR$, restricted to spheres meeting the ball. If the ball has radius r, we dominate its centre domain by $B(0, R + r)$. The total dominating mass is

$$p_* = \frac{8\pi F}{3L^4} \int_0^\infty R(R + r)^3 e^{-4R^2/L^2}\, dR.$$

For $r = 0.45$ and our parameters, $p_* \simeq 9.04762 \times 10^{-5} < 1$. With probability p_* we draw R from the normalised integrand and a centre uniformly from $B(0, R + r)$; otherwise the window is empty. Expanding $R(R + r)^3$ gives a four-component mixture in which R^2 has a gamma distribution, so no radius tail needs to be truncated. We reject a sphere that does not meet the ball. For an intersecting sphere we draw the Poisson count with mean $a(R)/(2R)$ times its intersection area, followed by independent uniform points on that spherical cap or whole sphere.

The correction component admits the same single-object method, using centres in $B(0, r + 1)$, their intensity κ, and the actual cluster mark law. The auxiliary rejection construction described below samples its marks without evaluating the marginal $\mathcal{M}q_3$ numerically. We flip the fair whole-law coin once for each independent window. This algorithm produces exact restrictions of the stated process, up to floating-point implementation; it does not construct several coupled windows from one whole-space realisation. For a general window one can instead sample the sphere band, shifts and finitely many relevant candidate spheres directly. The correction alternative can be sampled from Poisson parents on the window enlarged by $4M$, testing the $3M$ exclusion condition for parents within distance M of the window.

The measured correlations. Write $g_n = \rho^{(n)}/\lambda^n$. We anchor the first vertex at zero and let H_{nj} indicate a simplex-size bin, together with the condition that all pair distances are below 1. At order four we also require the unique circumsphere radius to be at most 2. Each reported bin has a positive lower edge. Our estimand is

$$\overline{g}_{nj} = \frac{\int H_{nj}(y)\rho^{(n)}(0, y)\, dy}{\lambda^n \int H_{nj}(y)\, dy}. \tag{5.1}$$

These restrictions make finite observables; they do not change the process or the domains of the theorem. The theoretical comparison is the bin average of $a_n/(\lambda^n \Delta_n)$, not its value at a bin midpoint. Anchored relative coordinates avoid window boundary loss: by stationarity, the numerator is the expected number of such ordered tuples whose first point lies in a unit-volume observation set, with the remaining points unrestricted. We use the prescribed λ throughout, since a spatial intensity estimate from one nonergodic realisation need not estimate this mixture mean.

Importance-sampled sphere tuples. Let $b(R) = a(R)/(2R)$ and $S(R) = \pi \min(1, 4R^2)$, the area of the cap of a sphere through zero lying in $B(0,1)$. Integrating the first point's surface measure over its possible centres gives

$$\int H_{nj}(y)\rho_Y^{(n)}(0,y)\,dy = \int_0^\infty 4\pi R^2\gamma(R)b(R)^n S(R)^{n-1}\mathbb{E}\{H_{nj}(0,X_2,\dots,X_n) \mid R\}\,dR. \quad (5.2)$$

Here the centre is Re, where e is uniform on the unit sphere, and the remaining points are independent uniform points in the cap. This identity comes directly from the conditional Poisson factorial measure, before its geometric density is evaluated. It integrates out the Poisson count using $\mathbb{E}(N)_n = m^n$. The sampled rows are factorial tuples, rather than ordinary Poisson realisations; unused background points are not generated.

We sample R with density

$$h_2(R) = \frac{4R}{L^2}e^{-2R^2/L^2}, \qquad h_3(R) = \frac{2}{L\sqrt{\pi}}e^{-R^2/L^2}.$$

The corresponding importance weights in (5.2) are $2A_2S(R)$ and $4\pi A_3 S(R)^2$. Both proposals retain the complete positive radius range. For four points the generating sphere is the tuple's unique circumsphere, so the observable's guard permits $0 < R \le 2$. With $Z = 1/6 + \log 4$, we use $h_4(R) = S(R)^3/(\pi^3 ZR)$ on this interval. Its weight is the constant $\pi^4 cZ/2$. All these weights follow from the generative intensity γ and surface intensity b; the reciprocal simplex formula is not inserted into the empirical numerator. Distances, areas and volumes are calculated from the sampled Cartesian coordinates.

Correction-cluster marks. We sample actual two- and three-point cluster marks by rejection, avoiding a numerical approximation to $F_2 = q_2 - \mathcal{M}q_3$. For $M = 1$, put $\varepsilon = 1/(2L)$ and $J = 2\varepsilon - \varepsilon^2$. Write $u = de$ and $v = te + r(\cos\phi\, e_1 + \sin\phi\, e_2)$ in a tangent basis. The integrable envelope

$$E(0,u,v) = \frac{2A_3}{dr}\min(1, \varepsilon^2/r^2)\mathbf{1}_{\{0<d<1,\ |t|<1,\ 0<r<1\}}$$

dominates q_3, and its mass is $T = 16\pi^2 A_3 J$. It is sampled by independent draws of d with density $2d$, uniform e, t and ϕ, and r with density $\min(1, \varepsilon^2/r^2)/J$. Acceptance with probability q_3/E produces the exact three-point mark law. Each accepted row contributes TH_{3j} to the third integral; its contribution to the second integral is $T/3$ times the sum of the three edge-bin indicators. Symmetry and re-rooting give the latter identity, and the three edge contributions are kept together when calculating variances.

For a pair, draw u from q_2/Q_2 by radial rejection, then draw an auxiliary v from the cylindrical conditional proposal $h(v \mid u)$ above. Accept the pair with probability

$$1 - \frac{q_3(0,u,v)}{q_2(0,u)h(v \mid u)}.$$

The subtracted quantity is at most $JL/2 \le 1/2$. Integrating over v gives the unnormalised accepted density F_2/Q_2; thus an accepted row contributes $Q_2 H_{2j}$. The auxiliary vertex is not an emitted cluster point. For every mark estimator we sum these contributions over all proposals and divide by the total proposal count. Rejected proposals contribute zero; the same complete rows enter the variance calculation. For example, with triple acceptance indicator I, the identity $\mathbb{E}[TIH_{3j}] = \int q_3(0,y)H_{3j}(y)\,dy$ gives the required unnormalised integral. The retained-parent intensity and the cluster size probabilities cancel the ordered-selection factors exactly as in the proof. We therefore need only sample marks for this local factorial experiment; we do not replace the dependent retained parents by independent ones.

For the ordinary-window implementation, one can avoid Q_3 even in the mark probabilities. At a retained centre, use candidate masses $(\lambda_C - Q_2)/\kappa$, $Q_2/(2\kappa)$, $T/(6\kappa)$ and $T/(2\kappa)$. These respectively propose a singleton, an accepted pair, an accepted triple, or an accepted triple proposal that emits only its centre. Rejections emit nothing. Their sum is at most one because $T \le Q_2/2$ and $\lambda_C \le \kappa$; their accepted masses are exactly the prescribed singleton, pair and triple mark probabilities.

Results and uncertainty. For each order we generated 524 288 independent sphere proposals. We also generated 524 288 pair proposals and 524 288 triple proposals, accepting 497 387 pair marks and 90 470 triple marks. Independent reference streams used 2 097 152 independent uniform draws $U_i \in B(0,1)^{n-1}$ per order. With $v_n = (4\pi/3)^{n-1}$ and reference sample size N, the estimators

$$\widehat{B}_{nj} = \frac{v_n}{N}\sum_{i=1}^{N} H_{nj}(U_i), \qquad \widehat{J}_{nj} = \frac{v_n}{N}\sum_{i=1}^{N} \frac{H_{nj}(U_i)}{\Delta_n(0,U_i)}$$

approximate the bin volume and its reciprocal-simplex integral. The summand in the second expression is set to zero outside the bin. The reference theory average is $a_n \widehat{J}_{nj}/(\lambda^n \widehat{B}_{nj})$; the empirical average uses the sampled factorial integral over $\lambda^n \widehat{B}_{nj}$. These reference draws are numerical integration data, separately labelled from the process samples. The full factorial integral estimates are one half the sphere estimate plus one half the relevant correction estimate, implementing the whole-law mixture without cross-component tuples.

All process contributions in this experiment are bounded, as are the reference integrands on the reported bins. Ordinary independent-sample standard errors therefore apply to these weighted observations, despite the original infinite fourth count moment. We propagate both the process sampling error and the independent reference integration error, retaining the covariance of the reference numerator and denominator. Error bars are 1.96 estimated Monte Carlo standard errors; they are approximate pointwise intervals, not finite-sample coverage guarantees.

The ratios of estimated to theoretical bin averages range from 0.9767 to 1.0043. Their standard errors range from 0.0019 to 0.0253; every reported deviation is below two standard errors. The correction contributes approximately 45–64% of the pair integrals and 17–51% of the triangle integrals in these bins. Its fourth contribution is zero by construction. Figure 1 and Table 1 report the normalised correlations; the supplement contains all bins, full precision and separate reference integration errors. This agreement checks the sampled construction and the local measure calculation in the stated domains. It does not establish the exact identities, global regularity, or infinite fourth moment, which are consequences of the proof.

The reproducible supplement contains the code, random seeds, every process tuple and its contribution, all ordinary-window counts and nonempty point clouds, and the reference sufficient statistics. A feasibility pilot was kept separate; it led us to end the reported fourth-order bins at volume 0.075, since larger volumes were rare under the proposal.

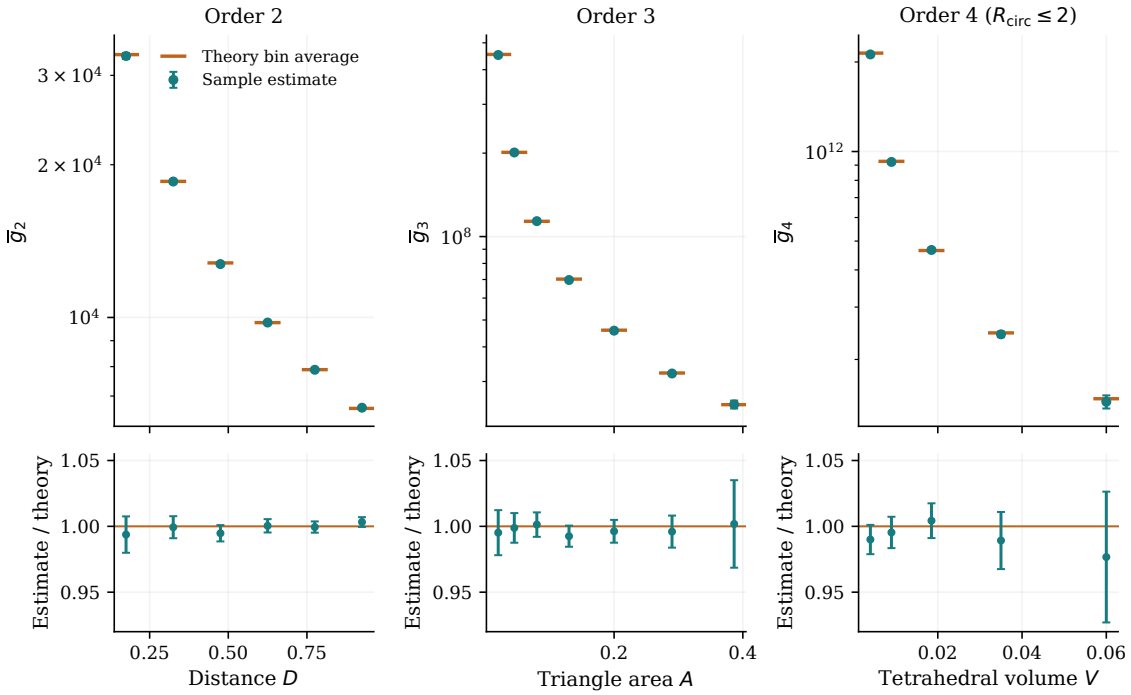


Figure 1: Bin-averaged g_2, g_3, g_4 from sampled factorial tuples and correction-cluster marks, with independent numerical integration of the theoretical bin averages. All bins have diameter below 1; order four additionally has $R_{\mathrm{circ}} \le 2$. The lower panels show the ratio to theory. Error bars include sampling and reference integration error and equal 1.96 Monte Carlo standard errors.

Table 1: Bin-averaged normalised correlations. The ratio column gives one Monte Carlo standard error in parentheses; the full-precision CSV also reports the separate error of the reference integration. Every bin has diameter below 1. At order four we additionally require $R_{\mathrm{circ}} \le 2$.

Order	Simplex bin	Estimate	Theory bin average	Ratio (SE)
2	$[0.1, 0.25)$	3.276×10^4	3.296×10^4	0.9937 (0.0070)
2	$[0.55, 0.7)$	9.772×10^3	9.768×10^3	1.0004 (0.0026)
2	$[0.85, 1)$	6.640×10^3	6.618×10^3	1.0033 (0.0019)
3	$[0.01, 0.03)$	4.519×10^8	4.540×10^8	0.9952 (0.0087)
3	$[0.1, 0.16)$	6.971×10^7	7.023×10^7	0.9925 (0.0041)
3	$[0.34, 0.433)$	2.479×10^7	2.475×10^7	1.0018 (0.0170)
4	$[0.002, 0.006)$	2.120×10^{12}	2.142×10^{12}	0.9900 (0.0057)
4	$[0.012, 0.025)$	4.667×10^{11}	4.647×10^{11}	1.0043 (0.0067)
4	$[0.045, 0.075)$	1.442×10^{11}	1.476×10^{11}	0.9767 (0.0253)

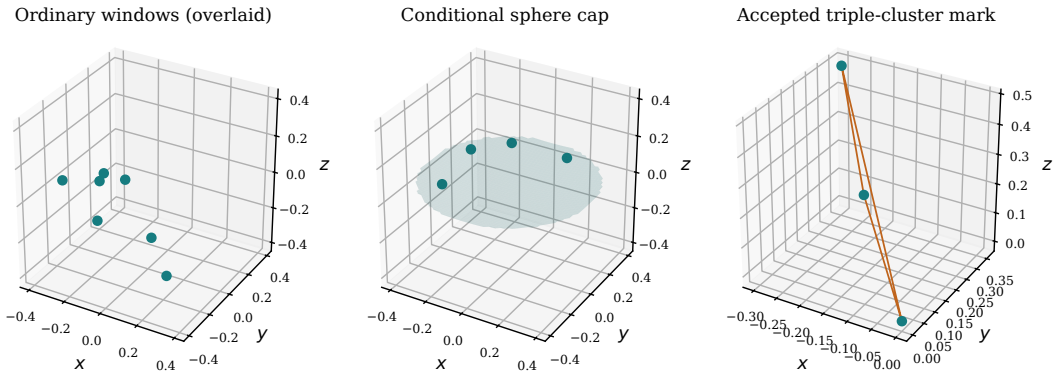


Figure 2: Saved point coordinates from three distinct kinds of sample. Left: the seven points from five million ordinary independent small-ball realisations are overlaid; they belong to seven different realisations. Centre: a separate conditional Poisson illustration, given a sphere of radius 2 centred at $(0, 0, 2)$ and at least four points in $B(0, 0.45)$. Right: one accepted three-point correction-cluster mark, before translation by its parent. The latter two illustrations are not ordinary draws from the whole-law mixture and are not added to the ordinary-window counts.

A Global densities and moments of the sphere process

This appendix proves Proposition 4.3. The local formulas do not suffice for that proposition: they describe tuples on one sphere in the prescribed diameter range, whereas the full measures also include tuples spread across several spheres and several centres. There are two distinct questions. We first prove absolute continuity for every sphere allocation. We then show that the resulting sum of densities is finite almost everywhere. At order four this second step needs a geometric argument, because the fourth measure is not locally finite.

A.1 Every allocation has an absolutely continuous measure

Conditionally on the sphere family, Y is Poisson with nonatomic locally finite directing measure (4.12). Hence $\alpha_Y^{(n)} = \mathbb{E}\Lambda^{\otimes n}$ as extended measures. Fix a band k and a prescribed allocation of n point labels to sphere indices (m_i, j_i), where $m_i \in \mathbb{Z}^3$ indexes a centre and $j_i \in \mathbb{Z}$ indexes a radius. Repeated sphere indices are allowed. A point in this allocation has the form

$$x_i = Z + P_k m_i + (U + j_i H)\omega_i, \qquad \omega_i \in S^2, \qquad i = 1, \dots, n. \tag{A.1}$$

The parameter measure is a nonnegative measurable weight times Lebesgue measure in (Z, U) and product surface area measure in the directions. Band membership, Poisson surface factors and the conditional retention probabilities are part of this weight. When a sphere index is repeated, its retention indicator is counted only once; the weight uses the retention probability for each distinct selected sphere. This changes no rank calculation below.

At positive radii, the angular derivatives of (A.1) span the two-dimensional plane $\omega_i^{\perp}$ in the ith position block. Together these give $2n$ independent tangent directions. Quotienting them out, the derivative with respect to (Z, U) has n rows

$$(\omega_i^{\mathsf{T}}, 1), \qquad i = 1, \dots, n. \tag{A.2}$$

For $n \leq 4$ these rows have rank n for almost every collection of directions under product area measure. At order two, rank failure requires the directions to coincide. Three distinct points on S^2 cannot be collinear, since a line intersects the sphere in at most two points. Thus three directions are almost surely affinely independent. Their affine plane intersects S^2 in a circle or a smaller set, of surface measure zero, so a fourth independently integrated direction almost surely avoids it. This proves the rank assertion, including all lower orders.

It follows that the map (A.1) has differential rank $3n$ almost everywhere. Here is the resulting absolute-continuity argument in detail. Use ordinary angular charts on the sphere. At a full-rank parameter point, select a nonzero $3n$-by-$3n$ minor of the derivative. The map consisting of the $3n$ position coordinates and the unused parameter coordinates is locally invertible. In these coordinates the position map is a coordinate projection. Fubini's theorem shows that the preimage of any Lebesgue-null position set has parameter measure zero. A countable cover by such inverse-function charts handles the full-rank set; the rank-failure set was already null for the parameter measure. Multiplication by a nonnegative measurable weight does not alter this conclusion. Thus the pushforward for the specified allocation is absolutely continuous.

There are countably many sphere allocations. Summing their measures, and then mixing over the countably many bands, proves global absolute continuity of $\alpha_Y^{(n)}$ for $n \leq 4$. In particular, no mass remains on a coincidence, collinearity or coplanarity set.

We can also exhibit the density without invoking a Radon–Nikodym theorem for a potentially non-locally-finite fourth measure. For each fixed band and bounded observation set, the number of influencing spheres has a deterministic finite bound. Moreover,

$$2\pi a(R)R \leq 2\pi b_k \sup_{0<R\leq b_k} a(R) < \infty.$$

The directing mass on that set is therefore bounded conditionally on the band, uniformly over the shift, phase and retention marks. All its count moments are finite. The conditional factorial measures for a fixed band are locally finite and absolutely continuous, so have finite-almost-everywhere densities $h_{n,k}$. The full measure has the extended density

$$\rho_Y^{(n)} = \sum_{k\geq 0} p_k h_{n,k}. \tag{A.3}$$

The empty alternative contributes zero. A countable sum of finite densities can still be infinite on a set of positive measure, so the remaining parts of this appendix prove that this does not occur.

A.2 Finite third count moments

Fix a ball $B = B(x, r)$. There is a constant $J_B < \infty$, independent of the selected band, bounding the number of candidate spheres that can meet B. Indeed, their centres must lie in $B(x, b_k + r)$, and the lattice spacing satisfies $P_k \geq 4b_k$ with $b_k \geq M$. A cubical packing bound therefore gives a uniform finite bound on the number of such centres, depending only on r/M. At any one centre, radii of spheres meeting B lie in an interval of length at most $2r$. A grid of spacing $H = 2M$ contains at most $\lfloor 2r/H \rfloor + 2$ such radii. Multiplying the two bounds gives J_B; retention can only decrease the number.

We also have the uniform area bound

$$\sigma_{z,R}(B) \leq C_B := 4\pi r^2. \tag{A.4}$$

For $R \le r$, use the full sphere area. For $R > r$, a nonempty intersection is a spherical cap. If $d = |z - x| > 0$, the cosine at its boundary is

$$\frac{R^2 + d^2 - r^2}{2Rd} = \frac{d + (R^2 - r^2)/d}{2R} \ge \frac{\sqrt{R^2 - r^2}}{R}.$$

Thus the cap area is at most

$$2\pi R^2 \left(1 - \sqrt{1 - r^2/R^2}\right) \le 2\pi r^2.$$

If $d = 0$ and $R > r$, the intersection is empty, so the bound holds there as well.

Write $m_{z,R} = a(R)\sigma_{z,R}(B)/(2R)$ for a sphere's contribution to $\Lambda(B)$. The number bound and (A.4) imply

$$\Lambda(B)^3 \le J_B^2 \sum_{(z,R)\ \text{retained}} m_{z,R}^3, \qquad \sigma_{z,R}(B)^3 \le C_B^2 \sigma_{z,R}(B).$$

Translation of the centre gives

$$\int_{\mathbb{R}^3} \sigma_{z,R}(B)\,\mathrm{d}z = 4\pi R^2 |B|. \tag{A.5}$$

Apply (4.9), (A.5) and Tonelli to obtain

$$\begin{aligned} \mathbb{E}\Lambda(B)^3 &\le \frac{\pi J_B^2 C_B^2 |B|}{2} \int_0^\infty \frac{\gamma(R) a(R)^3}{R}\,\mathrm{d}R \\ &= \frac{\pi J_B^2 C_B^2 |B|}{2}\, 2\int_0^\infty f(R^2)\,\mathrm{d}R = \frac{\pi J_B^2 C_B^2 |B|}{2} K\sqrt{\pi} < \infty. \end{aligned} \tag{A.6}$$

The cancellation $\gamma a^3/R = 2f(R^2)$ treats both small and large radii. For a Poisson variable of mean s, the third raw moment is $s^3 + 3s^2 + s$. Conditional expectation and (A.6) therefore give $\mathbb{E}Y(B)^3 < \infty$. Every bounded Borel set is contained in a ball, proving the third-moment claim. In particular, the full second and third measures are locally finite, so their densities in (A.3) are finite almost everywhere globally.

A.3 Finite fourth density away from a null exceptional set

The fourth measure has infinite mass on some bounded sets, so a moment bound cannot prove its density finite. Instead, we show that near almost every quadruple, sufficiently large radius bands make no contribution. Only finitely many locally finite fixed-band measures then remain.

Fix a noncoplanar quadruple $\mathbf{x} = (x_1, x_2, x_3, x_4)$, and translate x_1 to zero. Put $u_i = x_i - x_1$, $i = 2, 3, 4$, and let W be the invertible matrix with rows u_i^{T}. Let D_* denote the quadruple diameter. For all sufficiently large k, $2b_k + 2M > D_*$. By (4.10), the four points then cannot lie on spheres with two different centres. Any allocation in such a band must use concentric spheres.

Let their common centre, relative to x_1, be z, and let $R = |z|$ be the radius of the sphere through x_1. Because all candidate radii use the same phase and step H, the other three radii have the form $R + Hm_i$, where $m = (m_2, m_3, m_4) \in \mathbb{Z}^3$. The reverse triangle inequality gives

$$|Hm_i| \le |u_i| \le D_*. \tag{A.7}$$

The equations for the other points are $|u_i - z|^2 = (R + Hm_i)^2$. Subtracting $|z|^2 = R^2$ and putting $n = z/R$ gives

$$Wn = -Hm + \frac{b}{2R}, \qquad |n| = 1, \qquad b_i = |u_i|^2 - H^2 m_i^2. \tag{A.8}$$

Taking norms after multiplication by W^{-1} yields

$$R\left|1 - H|W^{-1}m|\right| \le \frac{1}{2}|W^{-1}b|. \tag{A.9}$$

This bounds R unless $H|W^{-1}m| = 1$.

For $m = 0$ the exceptional equality is impossible. For any fixed nonzero integer vector m, it is contained in the zero set of the polynomial

$$P_m(W) = H^2|\operatorname{adj}(W)m|^2 - (\det W)^2. \tag{A.10}$$

This polynomial is not identically zero: substitution $W = tI$ gives $t^4(H^2|m|^2 - t^2)$. A nonzero real polynomial has a Lebesgue-null zero set, as follows by induction on the number of variables and Fubini, using the finitely many roots of a nonzero one-variable polynomial. The coordinate change from the four vertices to (x_1, W) is invertible and linear. Consequently the union of all the sets $P_m(W) = 0$, together with the coplanarity set, is Lebesgue-null in the position space. Call this exceptional union $\mathcal{N}$.

Take a quadruple outside $\mathcal{N}$. Choose a bounded neighbourhood on which every diameter is at most D_0 and W^{-1} remains bounded. Only the finite integer set

$$\mathcal{I} = \{m \in \mathbb{Z}^3 : |Hm_i| \le D_0, i = 2, 3, 4\}$$

can appear in (A.7). At the chosen quadruple, the numbers $|1 - H|W^{-1}m||$, $m \in \mathcal{I}$, have a positive minimum. By continuity and finiteness of $\mathcal{I}$, shrink the neighbourhood so that all these numbers are at least a common $\varepsilon_0 > 0$. The quantities $|W^{-1}b|$ have a common finite bound there, again because $\mathcal{I}$ is finite. Equation (A.9) gives a uniform upper bound on R for every concentric allocation in that neighbourhood. The other radii differ from R by at most D_0, so they are bounded as well.

For all sufficiently large bands, different-centre allocations are excluded by (4.10), while concentric allocations are excluded because the band's lower radius endpoint exceeds this common bound. Thus, on the chosen bounded configuration neighbourhood, the full fourth measure is a sum over only finitely many bands. Every such fixed-band measure is locally finite by Section A.1. Its sum has a density finite almost everywhere on the neighbourhood.

The complement of $\mathcal{N}$ has a countable cover by such neighbourhoods, so the fourth density is finite almost everywhere there. The absolute-continuity proof in Section A.1 already shows that $\mathcal{N}$ itself has zero factorial measure. This finishes the proof of Proposition 4.3, both inside and outside the prescribed diameter domains.

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